

Part 1: Mechanical Design and Information Model

The RASCL robot features three axes and one motor dedicated to operating the end effector, also called *end of arm tooling (EOAT)*. Your goal is to design a gripper capable of handling small boxes. Currently, the end effector is incomplete and cannot yet perform this task.

As the purpose of this lab is to introduce you to the diversity of industrial automation and the skills required in this field, your task is to design a suitable end effector. You will use the 3D CAD software *onshape* for your design, which will then be 3D-printed using PETG material filament.

Since you may not hear design engineering courses in your studies so far, we will provide you with resources and tips to support your entry into 3D design. This will help you prepare your design:

- Explore the IRS ILIAS course “VSA - CAX” on design engineering to
 - learn about effective design strategies.
 - explore design principles optimized for 3D printing.
 - Introduction to 3D modeling in *onshape*, as a beginning some online resources, feel free to explore more
 - *onshape* Learning Center <https://learn.onshape.com/learn/learning-path/introduction-to-cad>
 - First 4 videos of this series:
https://www.youtube.com/watch?v=pMWnsHpDIQE&list=PLxmrkna-ixrlQmsPR3MITi4Ru1bnMH4-l&ab_channel=CADSessions
 - Get familiar with the FFF (Fused Filament Fabrication) printing process.
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To successfully complete your task, please adhere to the following **Design Requirements**:

- Design all end effector components using *onshape* and save them in your group folder.
- The number of parts is not limited, but fewer parts are preferable for simplicity and ease of assembly.
- Technical drawings are offered for the connection points of your design and the existing robot.
- You may use the following additional materials:
 - M3 and M4 hexagon socket head cap screws in standard lengths (according to DIN EN ISO 4762, see *Mechanical and Metal Trades Handbook / Tabellenbuch Metall*), M3 from length 6mm to 25mm and M4 from length 8mm to 20mm.
 - M3 and M4 hex nuts according to DIN EN ISO 4032 (DIN934)
 - A piece of raw PETG filament (diameter 1.75 mm) in the desired length from the filament spool.

Manufacturing Constraints

- The maximum length of PETG filament: **25m** (desired material can be seen in the slicer)
- The maximum printing time is limited to: **7h** (can be seen in the slicer)
- All parts must be printable on a **Prusa i3 MK3S** with a **0.4mm nozzle** and 0.2mm QUALITY (top right corner) and must fit on the printer's build plate (250W x 210D x 210H).
- The prints are executed on the *Prusa i3 MK3S* or the *Prusa XL* printer. Our workshop staff will decide which printer is chosen for your print. Please prepare for the print in the *Prusa Slicer* for the *Prusa i3 MK3S*.
- Please only use support material, where you think it is necessary. Please refrain from using the setting "everywhere".

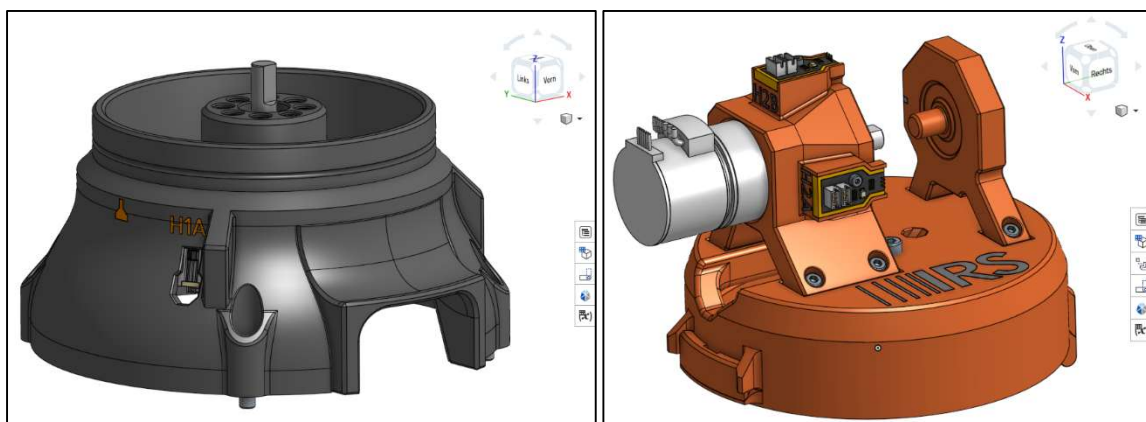
Printing Preparation (CAM = Computer Aided Manufacturing)

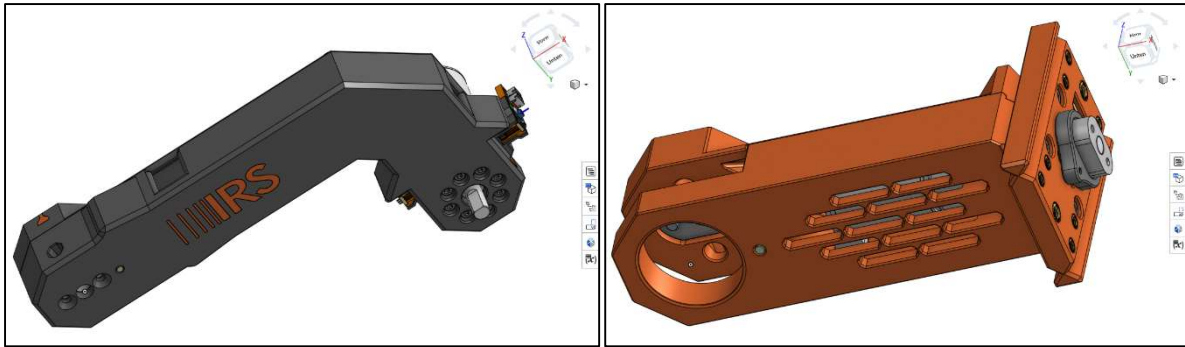
- You must prepare the print job using *PrusaSlicer*.
- Tip: Most default slicer settings are sufficient for this task.
- Choose the part orientation on the printing bed wisely.

URDF

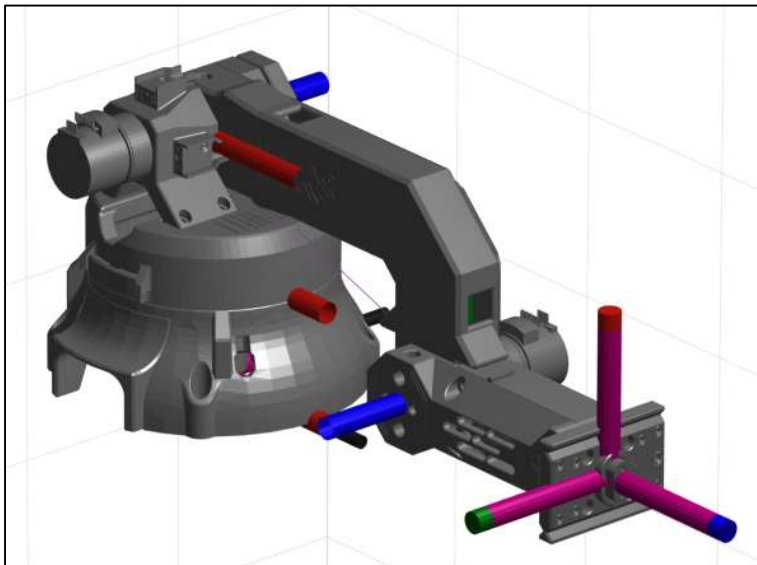
Later in this lab, your task will be to control the robot arm using ROS. To do this effectively, it is helpful to have an information model of the robot. A suitable format for this is the Unified Robot Description Format (URDF). Please familiarize yourself with URDF and create a description of the robot based on the technical drawings and CAD models (screenshots provided below). The URDF should include not only visual information but also self-collision data for the robot.

For better understanding, we have provided an urdf file for the base and shoulder. Note that the shoulder does not have a separate collision file, as its collision geometry is already included in the base file. Visual information is stored in dae files, while collision information is stored in stl files. Both file types for each axis share the same origin coordinate system. The origin coordinate systems for each axis can be seen in the screenshots below. All necessary files are available for download on ILIAS.



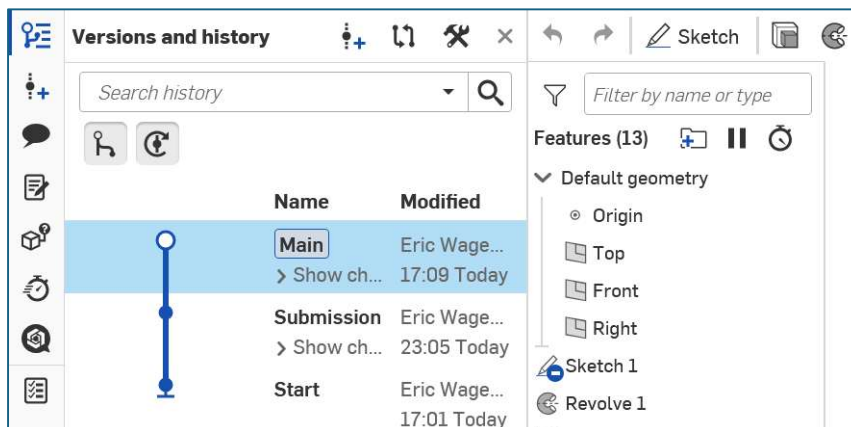


Your rendered urdf-file might look like this (here in Matlab, there is other software available):



Submission Guidelines

- Submit your files via ILIAS course in your group folder.
- Deadline: Sunday May 3rd, 2025, 23:59
- Deliverables:
 - Folder *URDF* containing:
 - `group<#number>UrdfModel.urdf` file
 - missing files your gripper design
 - Folder *Gripper* containing:
 - The `group<#number>GripperSubmission.3mf` file from *PrusaSlicer* with all your settings (this file will be used by the workshop staff for printing and is part of the evaluation).
 - A list `group<#number>Material.txt` of all additional components required for your end effector (e.g., screws, filament pieces) in a .txt document.
 - One page flyer (DIN A4) to promote your gripper → `group<#number>Flyer.pdf`
 - Please compress all files into a `group<#number>WP1.zip` file.
 - Create a version of your onshape document called *Submission*.



Additional printing

You might experience that your designed gripper is not working properly or breaks during your tests in later work packages. You can refine your design and get additional prints during the semester. You will have the chance of additional prints. Prints without the correct naming conventions or exceeding the limits mentioned above will not be printed. Improved designs are welcome but will not be included in the evaluation of your submission of the work package 1.

To get a print, please send your `group<#number>AdditionalPrint.3mf` to our workshop email address 3dprint-xl1@irs.kit.edu with *RASCL* as the subject. Your email must arrive before the following dates, such that they are printed during the following week and are available in the lab room in a bag. All print jobs you send will be executed; there is no option of cancelling your job. Please refrain from inquiring about the status of your print job. Submission dates:

- Sunday, June 7th 11:59pm
- Sunday, June 14th 11:59pm

- Sunday, June 21st 11:59pm
- Sunday, June 28th 11:59pm
- Sunday, July 5th 11:59pm
- Sunday, July 12th 11:59pm
- Tuesday, July 21st 11:59pm

All prints must be collected from the lab room before **July 31st**.

Evaluation guidelines

For evaluation we will focus on the following aspects:

- Functionality
 - Ability to grip a cube safely
 - Ease of assembly to the robot
 - Manufacturing
 - Short printing time
 - Material consumption of printer
 - Use of support materials
 - Amount of additional material
 - Creativity/originality of your design
 - What is your key selling point?
 - URDF
 - Running URDF file
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Consulting hour for work package 1

There will be consulting hours for this work package on Monday 27th of April at 3pm in lab 412 (building 30.33). Please prepare dedicated questions, only these are going to be answered in person. There will not be something like a lecture.